MovementLogger — Sensor & Module Catalog

8 Teile · 2026-05-21 10:08 UTC · BOM aus movement_logger_firmware · via crawl2pump

8 neu (7 Tage) 0 aktualisiert

Dev Boards 4

STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C)

OSS firmware · USB-C pluggable

☑ USB-C ☐ WiFi ☑ Bluetooth ☐ GPS ☑ Motion sensors ☑ SD-card · L×B×H 6.3 × 4 × 2 cm · 94 g

MCU: STM32U585 · Cortex-M33 @160 MHz · 2 MB flash / 786 KB SRAM · TrustZone · ultra-low-power

Datasheet: https://www.st.com/resource/en/user_manual/um3133-sensortilebox-pro-stmicroelectronics.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

neu **STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C) · STEVAL-MKBOXPRO @ DigiKey** — **CHF 94.56**

STMicroelectronics @ DigiKey · in stock

Primary target board. USB-C for charge + STM32 DFU flashing. · USB-C · firmware: open-source

https://www.digikey.com/en/products/detail/stmicroelectronics/STEVAL-MKBOXPRO/19721662

GPS / GNSS 14

GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA)

OSS firmware · SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card · L×B×H 4 × 4 × 1.3 cm · 80 g

Datasheet: https://www.sparkfun.com/products/14986

neu **GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) · GPS-14986 @ DigiKey** — **CHF 12.85**

SparkFun @ DigiKey · out of stock

Active GPS antenna (~28 dB LNA, 3 m RG-174 coax, SMA male). Magnetic base — sits on the board's non-metal top or the case lid for best sky view. Needs a U.FL → SMA pigtail to mate with the MAX-M10S breakout's U.FL connector — see next…

https://www.digikey.com/en/products/detail/sparkfun-electronics/14986/9682235

Interface Cable SMA → U.FL, 100 mm (pigtail)

OSS firmware · SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card · L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: https://www.sparkfun.com/products/9145

neu **Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com** — **USD 5.75**

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL ·…

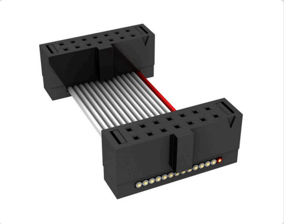
https://www.sparkfun.com/products/9145

Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm)

OSS firmware · SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card · L×B×H 10 × 0.5 × 0.2 cm · 3 g

Datasheet: https://suddendocs.samtec.com/catalog_english/ffsd.pdf



neu **Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm) · FFSD-07-D-04.00-01-N @ DigiKey** **CHF 9.21**

Samtec @ DigiKey · in stock

Solderless JP2 plug — 2×7 1.27 mm shrouded socket on both ends, mates with the STEVAL's FTSH-107 programming header. To break the ribbon's other end out to 0.1" DuPont (which slides onto the SparkFun MAX-M10S breakout's UART pins) you also…

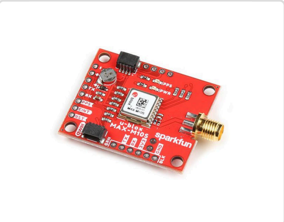
https://www.digikey.com/en/products/detail/samtec-inc/FFSD-07-D-04-00-01-N/6613350

SparkFun GPS Breakout - MAX-M10S (Qwiic) OSS firmware Qwiic / I²C

☐ USB-C ☐ WiFi ☐ Bluetooth ☑ GPS ☐ Motion sensors ☐ SD-card L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu **SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey** **CHF 35.80**

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. · Qwiic / I²C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314

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u-blox MAX-M10S GNSS module OSS firmware UART pins

☐ USB-C ☐ WiFi ☐ Bluetooth ☑ GPS ☐ Motion sensors ☐ SD-card L×B×H 1.01 × 0.97 × 0.25 cm · 0.60 g

Datasheet: https://content.u-blox.com/sites/default/files/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

neu **u-blox MAX-M10S GNSS module · MAX-M10S-00B @ DigiKey** **CHF 10.38**

u-blox @ DigiKey · in stock

Wired to UART4 @ 38400. The firmware's GPS source. · UART pins · firmware: open-source

https://www.digikey.com/en/products/detail/u-blox/MAX-M10S-00B/15712906

Enclosure 6

Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) OSS firmware enclosure

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 12 × 9 × 6 cm · 154 g

Datasheet: https://www.hammfg.com/electronics/small-case/plastic/1554.pdf

neu **Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) · 1554G2GYCL @ DigiKey** **CHF 22.77**

Hammond Manufacturing @ DigiKey · in stock

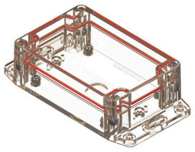
Polycarbonate IP66 project box, external 120 × 90 × 60 mm, clear PC lid. Lid sealed with O-ring gasket and four corner screws. RF-transparent so onboard GPS (and any radar) reads through the wall. The Hall sensor reads a passing magnet…

https://www.digikey.com/en/products/detail/hammond-manufacturing/1554G2GYCL/2359944

SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm)

OSS firmware | enclosure

☐ USB-C | ☐ WiFi | ☐ Bluetooth | ☐ GPS | ☐ Motion sensors | ☐ SD-card | L×B×H 12 × 8 × 4 cm · 75 g



neu **SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm) · RBF53P06C10C @ DigiKey**

CHF 21.97

SERPAC @ DigiKey · in stock

Polycarbonate IP67 project box, external 120 × 80 × 40 mm (4.72 × 3.15 × 1.59 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. IP67 = submersion to 1 m for 30 min, the right call for a foilboard…

https://www.digikey.com/en/products/detail/serpac/RBF53P06C10C/17100309

Solderless alternative — bring-up only

If you don't want to touch the iron yet (or you want to verify the wiring before committing to a permanent build), the BOM anchors a solderless 4-SKU assembly on one distributor-tracked Part (`samtec-ffsd-07-100mm`) that lets you plug into JP2 without soldering. **Caveat:** a vibrating pumpfoil board can wiggle the cable loose mid-session — use this for bring-up / development, switch to the soldered rig below for production.

Reference videos (watch first)

- ControllersTech — GPS Module and STM32 (NEO-6M): generic STM32 + u-blox GPS over UART. NEO-6M but the wiring is identical to MAX-M10S (TX↔RX crossover, GND, 3.3 V). Watch this one if you watch one.
- SparkFun — MAX-M10S Product Showcase: the specific GPS breakout from this BOM (PRT-18037). Shows the U.FL connector and the UART header pins you'll connect to.
- Chris Park — STM32 + u-blox UART NMEA, **wiring from 11:33**: the closest-fit hookup demo. Uses M8N (same pinout as the MAX-M10S), shows 4 DuPont female leads plugged onto VCC/GND/TX/RX on the GPS module. Translate directly to the STEVAL: JP4 → 3 V & GND, JP2 → TX/RX (crossover: STEVAL TX → GPS RX, STEVAL RX → GPS TX).

There is no exact "STEVAL-MKBOXPRO + solderless JTAG cable + MAX-M10S" video — that combination is bespoke to this build. The two videos above cover the load-bearing concepts (UART wiring + the specific GPS module).

Shopping list (4 SKUs, ~CHF 17 total)

#	Part	Where / Price
1	**Samtec FFSD-07-D-04.00-01-N** — 14-pin 1.27 mm IDC ribbon, 100 mm (the BOM card above)	DigiKey CH · ~CHF 9
2	**1.27 → 2.54 mm SWD adapter PCB** (no canonical Mouser/DigiKey MPN exists; survey of DigiKey + Mouser + Farnell + Distrelec + ChipDepot + Bastelgarage + Amazon DE found no Western-brand SKU for this category)	AliExpress · ~CHF 3 · search "Cortex SWD 14-pin 1.27 to 2.54 adapter"
3	**4× female-female DuPont jumpers**, ~10 cm	Bastelgarage · ~CHF 4 / 10 Stk
4	**2.54 mm 7-pin female header strip** — slides onto JP4 for the 3.3 V power tap	any electronics shop · ~CHF 1

Why this isn't a single SKU: a 14-pin 1.27 mm shrouded socket → loose 2.54 mm female DuPont cable is exactly what the buyer wants in one piece, but the category isn't stocked by any Western distributor (DigiKey, Mouser, Farnell, Distrelec, ChipDepot, Bastelgarage all checked); only generic Chinese OEM listings on Amazon/AliExpress fill it (~CHF 5–8, seller volatile). If you want to skip the adapter PCB + DuPonts, an AliExpress search for "14 Pin 1.27mm SWD to Dupont Cable" surfaces the single-cable version — but the BOM doesn't track it because no listing stays stable long enough.

Step 1 · JP4 power tap (3.3 V)

1. The JP4 connector is the 7-pin 2.54 mm strip on the side of the board (Multicomp MC-SVT1-S07-G).
2. **Set the JP4 domain switch to 3 V** (not 1.8 V).
3. Power on the box and **meter the JP4 3 V pin**: must read **3.25–3.40 V**. Power off again.
4. Slip a standard **2.54 mm 7-pin female header strip** onto JP4 (no solder — slides on). DuPont jumper from the 3 V pin → GPS VCC.

Step 2 · JP2 data lines (UART4)

1. Plug the JTAG cable into JP2 (FTSH-107 keyed/shrouded socket — only goes on one way).
2. Wire 3 DuPont jumpers from the adapter PCB's 2.54 mm side to the **SparkFun MAX-M10S breakout's UART header**:

STEVAL JP2 pin	Net (MCU)	→	SparkFun GPS pin
13	UART4_RX / PA1	→	**TX**
14	UART4_TX / PA0	→	**RX**
7 or 11	GND	→	**GND**

Note the TX/RX crossover (each end's "TX" pairs with the other's "RX"). The SparkFun breakout's UART header has male 2.54 mm pins populated — female DuPont leads slide on directly. **Do not touch SWD pins (4 / 6 / 8 / 10 / 12)** — they're the debug bus; bridging one means bricked debug access.

Step 3 · Bring-up test

1. Insert the SD card. Take the box **outdoors, clear sky**. Power on.
2. Cold fix is typically reached in **~30–60 s**. Confirm with a growing `GpsNNN.csv` on the SD card, or the BLE SensorStream `gps_valid` flag, or the antenna-test buzzer build.
3. No fix in a few minutes? **Swap pin 13 ↔ pin 14** — reversed TX/RX is by far the most common wiring mistake and swapping is harmless. Then re-check GND continuity and GPS VCC = 3.3 V under load.

When to switch to soldering: once bring-up works and you're ready for water trials. Vibration + spray + a loose cable will end a session early; the soldered rig below is the durable answer.

Soldering the u-blox MAX-M10S GPS to the STEVAL-MKBOXPRO

The MovementLogger firmware reads GPS over **UART4** on MCU pins **PA0 (TX) / PA1 (RX)** at **38400 baud, 8N1, 3.3 V logic**. The MAX-M10S is **not** on the box — it must be wired by hand to **JP2**, the 14-pin programming connector (Samtec FTSH-107-01-L-D, 1.27 mm pitch). Full source: GPS_SOLDERING.md.

Tools you need

- Fine-conical-tip iron, flux, magnification (10× loupe or microscope), tweezers, steady bench.
- **30 AWG** enamelled / wire-wrap wire (≤ 0.5 mm), four lengths.
- Multimeter (continuity + DC volts).
- Kapton tape + a dab of hot glue for strain relief.

JP2 pinout — pin 1 = square pad / silk triangle

Pin	Net	Use?
1	1V8 rail (JTAG ref)	**NOT** power — 1.8 V, weak
4 / 6 / 8 / 10 / 12	SWDIO / SWCLK / SWO / JTDI / NRST	Leave alone (debug bus)
7	GND	OK for GPS GND
11	GND	OK for GPS GND
13	**UART4_RX (PA1)**	**GPS TX → here**
14	**UART4_TX (PA0)**	**GPS RX → here**

Pins 3, 5, 9 carry no signal — leave unconnected. **Meter-verify** pin 1 against the silkscreen triangle and continuity-check every pin before soldering — connector orientation differs between board builds.

Wiring — note the TX/RX crossover

GPS pin	→	JP2 pin	Net (MCU)
TX (GPS out)	→	**13**	UART4_RX (PA1)
RX (GPS in)	→	**14**	UART4_TX (PA0)
GND	→	7 or 11 (meter-verified)	GND
VCC 3.3 V	→	**JP4 3 V pin** (not JP2!)	Board 3 V rail

GPS power — 3.3 V from JP4

- JP4 is the 7-pin extension-board connector (Multicomp MC-SVT1-S07-G). Set its domain switch to **3 V** (not 1.8 V).
- Power the box and **meter the JP4 3 V pin** — must read **3.25–3.40 V** before connecting GPS VCC. Power down again before soldering VCC.
- GPS GND and box GND must be common — take GND from JP4 or a JP2 GND pin.
- No graceful shutdown: the Hall-sensor magnet cuts the whole supply rail at once (firmware F-PWR-5). The on-SD format tolerates it.

Safety — read before the iron is hot

- **3.3 V logic only.** PA0/PA1 are **not 5 V tolerant.** A 5 V GPS breakout's TX into PA1 will destroy the MCU — level-shift or use a 3.3 V-native MAX-M10S carrier (e.g. the SparkFun breakout above).
- **Never bridge to the SWD pins** (4/6/8/10/12) — solder bridge → bricked debug access.
- Keep each pad heated **< 2 s**. FTSH-107 pads lift if cooked.
- **Strain-relieve all four wires** with Kapton + hot glue close to JP2 — a tugged wire rips the pad off the PCB.

Procedure

1. **Power down completely** — switch S1 off, USB unplugged, LiPo disconnected.
2. **Locate JP2** (14-pin fine-pitch header by the MCU). Identify pin 1.
3. **Continuity-check before soldering** (meter, board off): JP2 pin 13 ↔ MCU PA1, JP2 pin 14 ↔ MCU PA0, candidate GND pin ↔ battery (−). Confirm, don't assume.
4. **Pre-tin** the three JP2 pads and the wire ends with flux.
5. **Solder the data wires:** pin 13 ← GPS TX, pin 14 ← GPS RX. Then **GND**.
6. **GPS VCC last:** with the board powered, confirm 3.3 V at the JP4 3 V pin, power down again, solder the VCC wire.
7. **Inspect under magnification** for bridges — especially pin 14 to pin 12 and onto the SWD pins.
8. **Strain-relieve** all four wires (Kapton + hot glue).

Bring-up test

1. Insert the SD card, take the box **outdoors with clear sky**, power on.

2. The firmware sends the u-blox UBX config on boot, then expects 10 Hz NMEA. First fix is typically reached within **~30–60 s cold**.
3. Confirm a fix by **any** of: a growing `GpsNNN.csv` on the SD card · the BLE SensorStream `gps_valid` flag going set · (antenna-test build, `make GPS_ANTENNA_BEEP=1`) the buzzer beeping N times every 10 s with N = satellites.

No fix after several minutes outdoors? — in order of likelihood

1. **Swap the two data wires** (pin 13 ↔ pin 14). TX/RX reversed is the single most common wiring mistake and swapping is harmless to try.
2. Re-check **GND** continuity (cold/again) and that GPS VCC really is 3.3 V *at the module* under load.
3. Confirm the module is a 3.3 V part and actually powered (most MAX-M10S carriers have a power LED).
4. Give it a longer cold-start with an unobstructed sky view; indoors it will never fix.